

Capstone Proposal

Computer Science & Robotics | Good Hope Country Day School

Student Name

Project Title

Trimester

1. Project Description

Two or three sentences. What are you building?

2. Track

Pick one.

- ☐ Robotics & Competition
- ☐ Product & Fabrication
- ☐ Software
- ☐ Media & Documentation

3. Skills From Class

Check two or more. One is fine if you're pushing it somewhere genuinely hard. Explain what makes it hard in the box below.

- ☐ RoboMaster programming
- ☐ Micro:bit
- ☐ 3D printing and design
- ☐ Laser cutting
- ☐ AI chatbots (Gemini)
- ☐ Vibe coding
- ☐ Flutter
- ☐ Raspberry Pi web development
- ☐ An independent learning specialty (name it): _____
- ☐ Other: _____

If you checked only one skill, what makes it hard:

4. Tools, Technologies & Materials

What you'll build with, and any physical materials you need that aren't already in the lab (a specific sensor, a print that takes hours, a certain size of acrylic). Name it now so it can be ready before you start building.

5. Goal by Your Deadline

Be specific. What does "done" look like?

6. Why This Project?

Why you, why this.

7. Scope It

Sort your feature list into these four buckets. This is the part that keeps a project from running out of time.

Category	What it means	Your items
Must have	The smallest version that still counts as finished. If you only get this far, you pass.	
Should have	What makes it good, not just working.	
Could have	Nice if there's time. First thing added, last thing missed.	
Won't have this time	Name something you're deliberately skipping. Saying it now means nobody expects it later.	

8. Timeline

Work backward from your deadline. What has to be true at each point?

When	What has to be true
Final week (deadline)	
One week before	

When	What has to be true
Two weeks before	
Week 1 (right after approval)	

9. If You Run Out of Time

What's the first thing you cut, and what version still works without it?

10. Combining With Another Project? (optional)

If this also counts toward the Pi Custom Website project, explain how. Leave blank if not applicable.

How to Submit

Upload this filled template to Schoology as a PDF or DOCX. Not a shared link. The file itself.

For Teacher Use

- ☐ Approved
- ☐ Needs Revision
- ☐ Approved as Combo

Notes

Date

Example Proposal

Use this as a reference for how to fill out your own.

Student Name	Amara Petersen
Project Title	Reef Watch: a St. Croix water-temperature dashboard
Trimester	Fall 2026

1. Project Description

A Raspberry Pi with a temperature sensor that logs water temperature from a bucket sample and shows it on a live website, so my sailing team can track conditions over the season.

2. Track

Software

3. Skills From Class

Raspberry Pi web development, vibe coding (to speed up the dashboard's front end).

4. Tools, Technologies & Materials

Raspberry Pi, a waterproof temperature probe (need to order one, about \$8), Flask, HTML/CSS.

5. Goal by Your Deadline

A working dashboard that shows the current reading and a graph of the last 7 days, live on a page I can pull up on my phone at practice.

6. Why This Project?

I sail with the youth team and we always guess at water temperature. I want real numbers, and I want to get better at Flask.

7. Scope It

Category	Your items
Must have	Sensor reads a temperature. Page shows the current number.
Should have	A 7-day graph. Auto-refreshes without reloading.
Could have	Text alert if the water drops below a set temperature.
Won't have this time	Multiple sensors at different depths. A mobile app.

8. Timeline

When	What has to be true
Final week	Dashboard is live, graph works, tested on my phone.

When	What has to be true
One week before	Graph is working. Only polish left.
Two weeks before	Sensor reads reliably and the page shows a live number.
Week 1	Sensor ordered and arrives. Pi is set up and reading something.

9. If You Run Out of Time

Cut the graph first. A page that just shows the current temperature still solves the actual problem.